

Hamza's Robot Hand Desk Doohickey

This project controls an InMoov-based robotic hand using a **PocketBeagle**, an **Adafruit PCA9685 Servo Driver**, and a set of **potentiometers**. The system allows for real-time manual control of the robotic fingers, mapping analog inputs (potentiometers) to servo angles, and saves the hand's pose upon powering down.

Project Link: [Hamza's Robot Hand Desk Doohickey - Hackster.io](https://hackster.io/hamza/robot-hand-desk-doohickey)

Hardware Architecture

The system is built around the PocketBeagle, which reads analog voltage values from potentiometers and sends PWM signals via I2C to a servo driver.

Components

- **Controller:** PocketBeagle (or BeagleBone Black)
- **Servo Driver:** Adafruit PCA9685 16-Channel 12-bit PWM/Servo Driver
- **Actuators:** 5x MG996R or similar hobby servos (InMoov Hand assembly)
- **Inputs:** 5x 10kΩ Potentiometers
- **Power:** 5V 4A External Power Supply (for servos)

Wiring Diagram Overview

- **I2C Connection:** PocketBeagle I2C pins connected to PCA9685 (SCL/SDA).
- **Servos:** Connected to channels 11-15 on the PCA9685.
- **Potentiometers:** Connected to PocketBeagle Analog Input (AIN) pins.

Software Build Instructions

Prerequisites

1. **PocketBeagle Setup:** Ensure your PocketBeagle is running a standard Debian image and is connected to the internet
2. **Python 3:** The software is written in Python 3.

Dependencies

Install the required Adafruit libraries for BeagleBone IO and the PCA9685 driver. Run the following commands on your PocketBeagle:

```
sudo apt-get update
sudo apt-get install python3-pip
sudo pip3 install Adafruit_BBIO
sudo pip3 install Adafruit_PCA9685
```

Installation

Clone this repository or copy the servoController.py file to your PocketBeagle.

```
git clone [https://github.com/your-username/your-repo-name.git]
cd your-repo-name
chmod +x servoController.py
```

Software Operation

Configuration

Before running, verify the pin configuration in the `__main__` section of servoController.py. The default mapping is:

Finger	Potentiometer Pin (AIN)	Servo Channel (PCA9685)
Thumb	P1_19	11
Index	P1_21	12
Middle	P1_23	13
Ring	P1_25	14
Pinky	P1_27	15

You can also adjust the `SERVO_CAL` dictionary in the code to tune the minimum and maximum pulse widths for your specific motors.

Running the Controller

Execute the script

```
sudo python3 servoController.py
```

Usage

1. **Startup:** The hand will automatically move to its last saved position (loaded from `last_pose.json`) or a neutral position if no save file exists.
2. **Control:** Rotate the potentiometers to move the corresponding fingers. The software has

a tolerance filter to prevent jittering

3. **Shutdown:** Press Ctrl+C to exit the program. The current hand position will be saved to last_pose.json automatically, so that the hand doesn't snap next time you boot it.

License and copyright information are included in the python script